MADS

Multi-Agent Distributed System

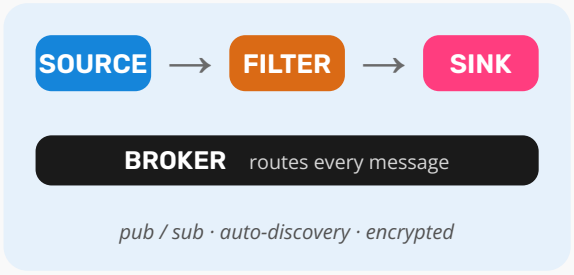
Connect sensors, devices & software into one distributed network.

Apache v2.0 · OPEN SOURCE

WHAT IS MADS?

MADS is a lightweight, cross-platform framework for building networks of **distributed agents** that exchange data over ZeroMQ. Wire up heterogeneous sensors, devices and software into one cooperating system — and start collecting and processing live data in **about one hour**.

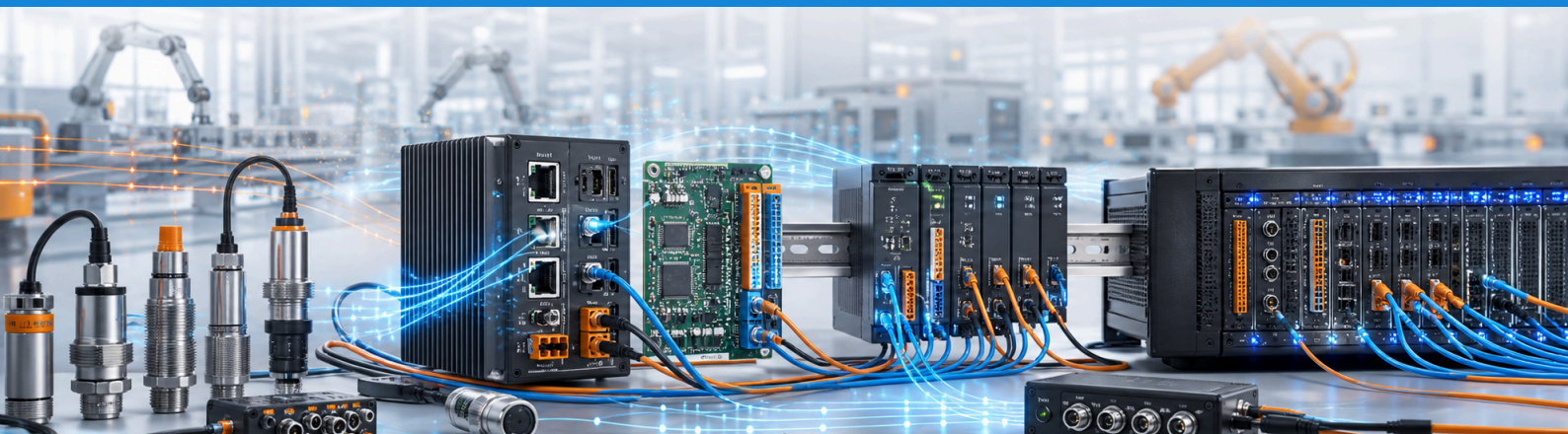
Every agent is a **source**, a **filter** or a **sink**. Customize behaviour with C++ plugins, or prototype in Python, R and Lua. No heavyweight middleware, minimal dependencies, tiny footprint.



WHY MADS?

- **Tiny footprint**
  ~20 MB installer with minimal dependencies — against ~750 MB for ROS 2.
- **Gentle learning curve**
  Productive in ~1 hour. The whole plugin API is barely 5 methods.
- **Runs everywhere**
  Windows, macOS, Linux — down to small boards like the Raspberry Pi.
- **Fast by design**
  ~150k messages/s (10 kB each) over the binary MsgPack protocol.
- **Truly extensible**
  Native C++/Rust plugins, or full agents scripted in Python, R and Lua.
- **Distributed & secure**
  Over-the-air plugin delivery and optional end-to-end encryption.

20 MB	1 h	150k	5	4+
INSTALLER SIZE	TO FIRST NETWORK	MSGS / SECOND	METHODS PLUGIN API	PLATFORMS



CONNECT ANYTHING

One network, every interface

A minimal core of general-purpose agents, extended by a growing ecosystem of plugins and agents at `github.com/MADS-Net`.

I/O & devices — Arduino · Arduino Uno Q · Bluetooth LE · Raspberry Pi GPIO · Serial

Languages — C++ plugins · Rust plugins · Python agent · Python library · R · Lua

AI & simulation — ONNX models · FMU / digital twins

Data & storage — HDF5 · MongoDB · CSV replay

Bridges & UI — MQTT · Text-to-speech · Terminal UI · Rerun live plots

Networking — Federation · Auto-discovery · OTA plugins

WHO'S USING MADS

University of Trento • MiRO Lab

Software stack of a **robotic walker** for neuro-rehabilitation of patients.

University of Brescia • MMTLab

Edge-AI cameras for real-time **3D human pose estimation**.

University of Perugia • Measurement Group

Data collection from **wearable sensors** and RFID transponders.

INSA Toulouse • Institut Clément Ader

Multi-sensor **energy-efficiency** monitoring on machining shopfloors.

TEACHING The new **MASSMa** course (Multi-Agent Systems for Smart Machining) runs jointly at the University of Trento and INSA Toulouse under the ECIU umbrella.

GET STARTED

- **GET MADS** `git.new/MADS`
- **DOCS** `mads-net.github.io`
- **SOURCE** `github.com/pbosetti/mads`
- **PLUGINS** `github.com/MADS-Net`
- **VS CODE** `MADSCode extension`

CONTACT

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MADS (v2.3 or later) is released under Apache 2.0 — free to use for any application.

